

📈 Business Impact & ROI Model

AgriAdvisor Local Fallback Engine (Small-Scale Deployment Projection)

⚙️ 1. Baseline Metrics & Core Assumptions (Monthly Run-Rate)

- 🔌 **System Drop Rate:** Approximately 800 queries drop per month due to rural network timeouts.
- 🔌 **Manual Recovery Cost:** Support agents must manually call back farmers to resolve dropped queries.
- 🔌 **Time & Labor:** Each manual callback/investigation takes an average of 15 minutes.
- 🔌 **Labor Rate:** Fully loaded cost of a tier-1 support agent is ₹ 150 per hour.
- 🔌 **Fallback Efficiency:** The local offline reasoning engine successfully answers 80% of dropped queries natively.
- 🔌 **Customer Value:** Average Revenue Per User (ARPU) is ₹ 50/month. Unanswered drops have a 20% churn risk.

🕒 2. Operational Cost Savings (Time & Labor)

Metric	Calculation Logic	Impact
Total Manual Time Avoided	800 drops × 80% handled locally × 15 mins per resolution	160 Hours / mo
Labor Cost Reduced	160 hours saved × ₹ 150 per hour	₹ 24,000 / mo
API Costs Avoided	640 fallback queries rerouted from cloud LLM	Negligible

🛡️ 3. Revenue Retention & Risk Mitigation

Metric	Calculation Logic	Impact
Churn Events Prevented	640 local resolutions × 20% risk of churn if left unanswered	128 Users Retained
Monthly Revenue Protected	128 retained users × ₹ 50 ARPU	₹ 6,400 / mo

🏆 4. Total Quantified Business Impact

Total Monthly Impact ₹ 30,400	Projected Annual ROI ₹ 364,800	Support Capacity Freed 1,920 Hours/yr (1 Full-Time Agent)
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💖 5. Qualitative & Strategic Benefits

- ✅ **Farmer Trust & Reliability:** Providing instant answers even in zero-network zones builds high brand loyalty and reliance on the AgriAdvisor platform.
- ✅ **Support Team Mental Bandwidth:** Removes highly repetitive, low-tier follow-up calls from the support queue, allowing agents to focus on complex agronomy escalations.
- ✅ **Graceful Degradation:** System degrades from "Cloud Intelligence" to "Local Intelligence" rather than showing an error screen, vastly improving the UX.